Example (Unbalanced dataset):

We have 100 persons.

- Label (1) refers to: has cold/flu
- Label (0) refers to: has NOT cold/flu

Actual data:

90: have not cold/flu 10: have cold/flu

Predicted data:

95: have not cold/flu 5: have cold/flu

Confusion matrix:

TP = 5	**FN = 5**
FP = 0	**TN = 90**

Accuracy:

Accuracy = (5 + 90) / (5+90+0+5) = 95 / 100 = 95%

Recall:

Recall = 5 / (5 + 5) = 5 / 10 = 50%

Precision:

Precision = 5 / (5 + 0) = 100%

Another Example:

Given the following table, suppose that **1** represents **positive**, and **0** represents **negative**:

Output of Actual data	Output of Predicted data
1	**1**
1	**0**
0	**0**
1	**1**
0	**0**
0	**0**
1	**1**
0	**1**
0	**1**
0	**0**

1. Find the confusion matrix
2. Find the accuracy
3. Find the precision
4. Find the recall

Answer:

- We will firstly find TP, TN, FP, and FN.

  TP = 3 TN = 4 FP = 2 FN = 1

- Confusion matrix =

3	**1**
2	**4**

- Accuracy = (3 + 4) / (3+4+2+1) = 7 / 10 = 70%
- Precision = 3 / (3+2) = 3 / 5 = 60%
- Recall = 3 / (3+1) = 3/4 = 75%